$$\begin{aligned}\lim_{x\to 2}(4x^2+3) &= \lim_{x\to 2}4x^2+\lim_{x\to 2}3 = 4\lim_{x\to 2}x^2+\lim_{x\to 2}3\\ &= 4\left(\lim_{x\to 2}x\right)^2+\lim_{x\to 2}3 = 4(2)^2+3 = 19.\end{aligned}$$

2. $\lim_{x\to 3}\left[\frac{x-2}{x+2}\right]$

Sol

$$\lim_{x\to 3}\left[\frac{x-2}{x+2}\right] = \frac{\lim_{x\to 3}(x-2)}{\lim_{x\to 3}(x+2)} = \frac{\lim_{x\to 3}x-\lim_{x\to 3}2}{\lim_{x\to 3}x+\lim_{x\to 3}2} = \frac{3-2}{3+2} = \frac{1}{5}$$

3. $\lim_{x\to 4}\sqrt{25-x^2}$

Sol

$$\lim_{x\to 4}(25-x^2)^{\frac{1}{2}} = \left(\lim_{x\to 4}25-\lim_{x\to 4}x^2\right)^{\frac{1}{2}} = \left(\lim_{x\to 4}25-\left(\lim_{x\to 4}x\right)^2\right)^{\frac{1}{2}} = \left(25-(4)^2\right)^{\frac{1}{2}} = 3$$

4. $\lim_{x\to 1}(3x^2-1)(9x+2)$

Example 1.4.6 *Evaluate the following limits:*

1. $\lim_{x\to 1}(x+4-x^2)$

Sol

2. $\lim_{x\to 3}\frac{3x+1}{x+2}$

Sol

3. $\lim_{x\to 1}(x+4)(x-1)$

Sol

4. $\lim_{x\to 2}\frac{x+2}{x-2}$

Sol

5. $\lim_{x\to\infty} \frac{3+\frac{1}{x}}{2+x}$

Sol

6. $\lim_{x\to\infty} \frac{4x+1}{x+9}$

Sol

7. $\lim_{x\to -1} \frac{x^2-1}{x+1}$

Sol

8. $\lim_{x\to -1} \frac{x^2+3x+2}{x+1}$

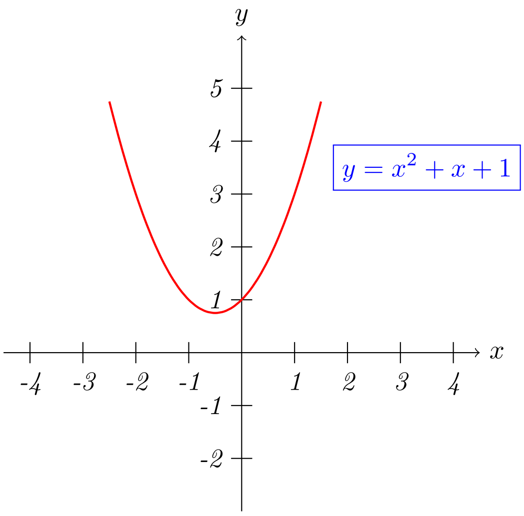
Sol

9. $\lim_{x\to 2} \frac{x^5-32}{x^2-4}$

Sol

10. $\lim_{x\to 0} \frac{x+\sin x}{\tan x+x}$

Sol



3. $\lim\limits_{x \to -1} \dfrac{x+1}{x^2+4x+3}$

Sol

$$\lim_{x \to -1} \frac{x+1}{x^2+4x+3} = \lim_{x \to -1} \frac{x+1}{(x+1)(x+3)} = \lim_{x \to -1} \frac{1}{(x+3)} = \frac{1}{-1+3} = \frac{1}{2}$$

4. $\lim\limits_{x \to 5} \dfrac{x-5}{\sqrt{x-3}-\sqrt{2}}$

Sol

$$\lim_{x \to 5} \frac{x-5}{\sqrt{x-3}-\sqrt{2}} . \frac{\sqrt{x-3}+\sqrt{2}}{\sqrt{x-3}+\sqrt{2}} = \lim_{x \to 5} \frac{(x-5)\left(\sqrt{x-3}+\sqrt{2}\right)}{(x-5)} = \lim_{x \to 5} \left(\sqrt{x-3}+\sqrt{2}\right) = \sqrt{2}+\sqrt{2} = 2\sqrt{2}.$$

5. $\lim\limits_{x \to 0} \dfrac{(1+x)^2-1}{x}$

Sol

6. $\lim\limits_{x \to 1} \dfrac{x^6-1}{x^2-1}$

Sol

7. $\lim\limits_{x \to 3} \dfrac{\frac{1}{x}-\frac{1}{3}}{x^2-9}$

Sol

8. $\lim\limits_{x\to 4} \dfrac{\sqrt{x}-2}{x-4}$

Sol

Example 1.5.8 *Evaluate the following limits.*

1. $\lim\limits_{x\to\infty} \dfrac{9x^2}{2x^2+4}$

Sol

By direct substitution, we get:

$$\lim_{x\to\infty} \frac{9x^2}{2x^2+4} = \frac{9(\infty)^2}{2(\infty)^2+4} = \frac{\infty}{\infty} \qquad ,(\textit{undetermined value})$$

$$\lim_{x\to\infty} \underbrace{\frac{9x^2}{2x^2+4}}_{f(x)} = \lim_{x\to\infty}\left(\frac{\cancel{x^2}[9]}{\cancel{x^2}\left[2+\dfrac{4}{x^2}\right]}\right) = \lim_{x\to\infty}\underbrace{\left(\frac{9}{2+\dfrac{4}{x^2}}\right)}_{g(x)} = \frac{9}{2+\dfrac{4}{\infty^2}} = \frac{9}{2+0} = \frac{9}{2}$$

2. $\lim\limits_{x\to\infty} \dfrac{x^3-x^2+2}{2x^2+x+1}$

Sol

$$\lim_{x\to\infty} \underbrace{\frac{x^3-x^2+2}{2x^2+x+1}}_{f(x)} = \lim_{x\to\infty} \frac{\cancel{x^3}\left(1-\dfrac{1}{x}+\dfrac{2}{x^3}\right)}{\cancel{x^3}\left(\dfrac{2}{x}+\dfrac{1}{x^2}+\dfrac{1}{x^3}\right)} = \lim_{x\to\infty}\underbrace{\frac{\left(1-\dfrac{1}{x}+\dfrac{2}{x^3}\right)}{\left(\dfrac{2}{x}+\dfrac{1}{x^2}+\dfrac{1}{x^3}\right)}}_{g(x)} = \frac{\left(1-\dfrac{1}{\infty}+\dfrac{2}{\infty^3}\right)}{\left(\dfrac{2}{\infty}+\dfrac{1}{\infty^2}+\dfrac{1}{\infty^3}\right)} = \infty$$

3. $\lim\limits_{x\to\infty} \dfrac{x(x+1)}{x^3+1}$

Sol

4. $\lim_{x\to\infty} \frac{6x+1}{\sqrt{4x^2+3}}$

Sol

5. $\lim_{x\to\infty} \frac{(x-3)(2x+1)}{2x^2-x+1}$

Sol

6. $\lim_{x\to\infty} \frac{(3x+4)(x-2)}{x(2x+1)(x+2)}$

Sol

7. $\lim_{x\to\infty} \frac{\sqrt{9x^2+5}}{x+3}$

Sol

1.6 Cauchy's Theorem

Theorem 1.6.4 ***Cauchy's Theorem:***

$$\boxed{\lim_{x\to a} \frac{x^n-a^n}{x-a} = na^{n-1}}$$

proof

3. $\lim\limits_{x\to\infty}\left(\frac{x+3}{x-2}\right)^x$

Sol

Let $u = x-2 \Rightarrow x = u+2 \Rightarrow (u\to\infty)$ *as* $(x\to\infty)$

$$\Rightarrow \lim_{x\to\infty}\left(\frac{x+3}{x-2}\right)^x = \lim_{u\to\infty}\left(\frac{u+2+3}{u}\right)^{u+2} = \lim_{u\to\infty}\left(1+\frac{5}{u}\right)^{u+2} = \lim_{u\to\infty}\left(1+\frac{5}{u}\right)^{u} . \lim_{u\to\infty}\left(1+\frac{5}{u}\right)^{2} = e^5$$

Exercise 1.10.1 *Evaluate the following limits.*

1. $\lim\limits_{x\to 3}\frac{x^2-2x-3}{x-3}$

Sol

2. $\lim\limits_{x\to 4}\frac{x^2-16}{4-x}$

Sol

3. $\lim\limits_{x\to b}\frac{(x-b)^{50}-x+b}{x-b}$

Sol

4. $\lim\limits_{h\to 0}\frac{\frac{1}{5+h}-\frac{1}{5}}{h}$

Sol

5. $\lim\limits_{w\to 1}\left(\frac{1}{w^2-w}-\frac{1}{w-1}\right)$

Sol

6. $\lim\limits_{x\to\infty} \dfrac{x^2+2x+9}{3x^3+x+1}$

Sol

7. $\lim\limits_{x\to 1} \dfrac{\sqrt{9x^2+7}}{3x+1}$

Sol

8. $\lim\limits_{n\to\infty} \left(\dfrac{1+2+3+\cdots+n}{n^2}\right)$

Sol

9. $\lim\limits_{x\to 1} \dfrac{x^3+x^2+5x-5}{x-1}$

Sol

10. $\lim\limits_{x\to 0} \left(\dfrac{\dfrac{1}{x-1}+\dfrac{1}{x+1}}{x}\right)$

Sol

11. $\lim\limits_{x\to 2} \dfrac{x^3(x-1)^6-8}{x-2}$

Sol

12. $\lim\limits_{x\to 2} \dfrac{\sqrt{x^2+12}-4}{x-2}$

Sol

13. $\lim_{x \to 1} \dfrac{x^{-1} - 1}{x - 1}$

Sol

14. $\lim_{x \to 4} \dfrac{4x^2 - x^2}{2 - \sqrt{x}}$

Sol

15. $\lim_{x \to 4} \dfrac{4 - x}{\sqrt{x^2 + 9} - 5}$

Sol

16. $\lim_{y \to 8} \dfrac{\sqrt[3]{y} - 2}{y\sqrt[3]{y} - 16}$

Sol

17. $\lim_{y \to 3} \dfrac{y^4 - 81}{y^2 - 9}$

Sol

18. $\lim_{\Delta x \to 0} \dfrac{(x + \Delta x)^3 - x^3}{\Delta x}$

Sol

19. $\lim_{x \to -2} \frac{x^5 + 32}{x^3 + 8}$

Sol

20. $\lim_{x \to 0} \frac{4x + \sin 3x}{x}$

Sol

21. $\lim_{x \to 0} \frac{\sin 3x}{\sin 9x}$

Sol

22. $\lim_{x \to 0} \frac{\sin 2x}{x^2 + 3x}$

Sol

23. $\lim_{x \to 2} \frac{\sin(x-2)}{x^2 - 4}$

Sol

24. $\lim_{x \to 0} \frac{1 - \cos 4x}{x^2}$

Sol

25. $\lim_{x \to 0} \frac{\tan 2x + 4x}{x + \sin x + \tan x}$

Sol

26. $\lim_{x\to 0} \frac{\cos 4x - \cos 2x}{x^2}$

Sol

27. $\lim_{x\to \infty} x \sin \frac{6}{x}$

Sol

2.

$$\lim_{x\to 0^+} f(x) = \lim_{\epsilon\to 0} f(a+\epsilon) = \lim_{\epsilon\to 0} \frac{\sin(\pi(0+\epsilon))}{0+\epsilon+1} = \lim_{\epsilon\to 0} \frac{\sin \pi\epsilon}{\epsilon+1} = \frac{\sin 0}{0+1} = \frac{0}{1} = 0$$
$$\lim_{x\to 0^-} f(x) = \lim_{\epsilon\to 0} f(a-\epsilon) = \lim_{\epsilon\to 0} \frac{\sin(\pi(0-\epsilon))}{0-\epsilon} = \lim_{\epsilon\to 0} \frac{\sin \pi\epsilon}{\epsilon} = \pi$$

$\Rightarrow \lim_{x\to 0} f(x)$ *not exist because* $\lim_{x\to 0^+} f(x) \neq \lim_{x\to 0^-} f(x)$

Example 1.2.7 *Find the constant such that the function is continuous on the entire real line.*

1.

$$f(x) = \begin{cases} 3x^2 & , x \geq 1 \\ ax-4 & , x < 1 \end{cases}$$

2.

$$f(x) = \begin{cases} \dfrac{4\sin x}{x} & , x < 0 \\ a-2x & , x \geq 0 \end{cases}$$

3.

$$f(x) = \begin{cases} 1-4x & , x < 2 \\ ax^2-3x+2 & , x \geq 2 \end{cases}$$

1.3 Properties of Continuity

Theorem 1.3.1 *If b is a real number and f and g are continuous at $x = c$, then the following functions are also continuous at c*

1. *Scalar multiple: bf*
2. *Sum or Difference: $f \pm g$*
3. *Product: fg*
4. *Quotient: $\dfrac{f}{g}$, if $g(c) \neq 0$.*

Note: The following types of functions are continuous at every point in their domains.

(i) Polynomials

(ii) Rational $r(x) = \dfrac{p(x)}{q(x)} \qquad , q(x) \neq 0$

(iii) Radical: $f(x) = \sqrt[n]{x}$

(iv) Trigonometric: $\sin x, \cos x, \tan x, \cot x, \sec x, \csc x$

Exercise 1.3.1 *Find the following:*

1. *Prove that* $f(x) = \sin(ax) \quad , (a \neq 0)$ *is continuous on* $\mathbb{R}$.

2. *Prove that* $f(x) = x^3 - x$ *is continuous on* $\mathbb{R}$.

3. *Discuss the continuity of the function* $f(x) = \ln(x^2 + 1)$

4. *Determine whether the following functions are continuous at the given points.*

   (a) $f(x) = \begin{cases} x^x - 1 & , x \neq 0 \\ 1 & , x = 0 \end{cases}$ *at* $x = 0$

   (b) $f(x) = \begin{cases} \dfrac{sin(\pi x)}{1 - x} & , x < 1 \\ \pi x^2 & , x \geq 1 \end{cases}$ *at* $x = 1$

Theorem 2.4.4 ***Quotient Rule***
If f and g are differentiable at x , $g(x) \neq 0$, then the derivative of $\frac{f}{g}$ at x exists and

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f^{'}(x) - g^{'}(x)f(x)}{(g(x))^2}$$

Proof

$$\begin{aligned}
\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] &= \lim_{h\to 0}\frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} = \lim_{h\to 0}\frac{g(x)f(x+h) - f(x)g(x+h)}{h(g(x)g(x+h))} \\
&= \lim_{h\to 0}\frac{g(x)f(x+h) - f(x)g(x+h) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{h(g(x)g(x+h))} \\
&= \frac{\lim\limits_{h\to 0}\frac{g(x)[f(x+h)-f(x)]}{h} - \lim\limits_{h\to 0}\frac{f(x)[g(x+h)-g(x)]}{h}}{\lim\limits_{h\to 0}(g(x)g(x+h))} \\
&= \frac{g(x)\lim\limits_{h\to 0}\frac{[f(x+h)-f(x)]}{h} - f(x)\lim\limits_{h\to 0}\frac{[g(x+h)-g(x)]}{h}}{\lim\limits_{h\to 0}(g(x)g(x+h))} \\
&= \frac{g(x)f^{'}(x) - g^{'}(x)f(x)}{(g(x))^2}
\end{aligned}$$

Note: $\lim\limits_{h\to 0} g(x+h) = g(x)$ because g is given differentiable and therefore is continuous.

5. **The Chain Rule**

Theorem 2.4.5 *If $y = f(u)$ is a differentiable function of u, and $u = g(x)$ is a differentiable function of x, then $y = f(g(x))$ is a differentiable function of x and*

$$\boxed{\frac{d}{dx}[f(g(x))] = f^{'}(g(x)).g^{'}(x)}$$

Example 2.4.15 *Evaluate $\frac{dy}{dx}$ for the following functions.*

1. $y = 4x^3 + \frac{3}{x} + 2\sqrt{x} - 7$

Sol

2. $y = 3\cos x - \frac{1}{2}\sin x$

Sol

3. $y = x^7 + \frac{1}{3}x^{-2} - \frac{1}{\sqrt{x}} + \sqrt{x^3}$

Sol

4. $y = x \sin x$

Sol

5. $y = x^3 \cos x + x^2 - 5$

Sol

6. $y = (x+4)(x^2 - x + 5)$

Sol

7. $y = \dfrac{3x^2+1}{6x+5}$

Sol

8. $y = \dfrac{1+\cos x}{1-\sin x}$

Sol

9. $y = 2^{-x} + 3^{x+1}$

Sol

10. $y = \log_4(x) - 2\ln x$

Sol

11. $y = x\sin(3x+4)$

Sol

12. $y = \dfrac{\sin x + \cos x}{\sin x - \cos x}$

Sol

13. $y = \log_3\left(x + \sqrt{x^2+1}\right)$

Sol

Exercise 2.4.2 *1. From the first principles, evaluate the first derivative for the following function.*

(i) $y = x^3$

(ii) $y = \dfrac{x}{x^2-5}$

(iii) $y = \sin 2x$

(iv) $y = xe^{-x}$

2. *Find* $\dfrac{dy}{dx}$ *for the following:*

(a) $y = \sin(x \cos 2x)$

(b) $y = \dfrac{x^2 + ax + b}{x^2 - ax + b}$

(c) $y = \sqrt{1 + \sqrt{2^x - 1}}$

(d) $y = \sin^2\left(\dfrac{2x}{x+1}\right)$

(e) $y = \dfrac{1 + 2\tan^2 x}{1 - \tan^2 x}$

(f) $y = \ln(\sec x + \tan x)$

(g) $y = \dfrac{1}{\sqrt{1+\sqrt{x}}}$

(h) $y = \dfrac{x^2}{\sqrt{a^2 - x^2}}$

(i) $y = xe^{-x^2}$

(j) $y = x\cos(\sin 2x)$

Sol

2.

$$y = \tan^{-1}\left(\frac{x}{a}\right) + 2\sin^{-1}\left(\frac{a}{x}\right)$$

Sol

3.

$$y = \frac{\cos^{-1} x^2}{x+1}$$

Sol

4.

$$y = e^{x^2} \sin^{-1} x$$

Sol

5.

$$y = \sin^{-1} 2x + \cos^{-1} 2x$$

Sol

6.

$$y = x\cos^{-1} x - \sqrt{1-x^2}$$

Sol

7.

$$y = \frac{1}{2}\left[x\sqrt{4-x^2} + 4\sin^{-1}\left(\frac{x}{2}\right)\right]$$

Sol

8.

$$y = \cos x . \sec^{-1} x$$

Sol

9.

$$y = e^{\sin^{-1} x} + \frac{1}{\sqrt{1+x^2}}$$

Sol

10.

$$y = \csc\left[\frac{x}{x+1}\right] - \sec^{-1}\left[x + \frac{1}{x}\right]$$

Sol

11.

$$y = \frac{\cot^{-1} x}{1 + x^2}$$

Sol

12.

$$y = \ln\left(x - \tan x\right)$$

Sol

13.

$$y = \csc^{-1}\left[\frac{1 - x}{x + 1}\right]$$

Sol

14.

$$y = \ln\sqrt{\frac{x + 1}{x - 1}} + \tan^{-1} 2x$$

Sol

(f) $x = 2^y - 2^{-y}$

Sol

Differentiate both sides directly:

$$1 = 2^y \ln 2 \, \frac{dy}{dx} + 2^{-y} \ln 2 \, \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{(2^y + 2^{-y}) \ln 2}$$

Exercise 1.2.1 *1. Find* $\frac{dy}{dx}$ *for the following:*

(i) $y = \pi^x - x^\pi$

Sol

(ii) $y = e^{\sin 3x}$

Sol

(iii) $y = \ln(x^2 - 1)$

Sol

(iv) $y = \ln\left(\frac{2^x - x}{2^x + x}\right)$

Sol

(v) $y = \log_3\left(\frac{x}{2^x - x}\right)$

Sol

(vi) $y = \tan^{-1}\left(e^{-x^2}\right)$

Sol

(vii) $y = \sin\sqrt{e^x - 1}$

Sol

(viii) $y = \dfrac{\ln(x) + 1}{\ln(x) - 1}$

Sol

(ix) $y = \dfrac{(x+1)^3\sqrt{x^2+4}}{\sqrt[3]{x+2}}$

Sol

(x) $y = \ln\left[\ln\left(\dfrac{x+1}{x-1}\right)\right]$

Sol

(xi) $y = \dfrac{\sin^4(x)\cos^3(x)}{\sqrt{x}}$

Sol

(xii) $xy = 2^{x-y}$

Sol

(xiii) $y + x = y^x$

Sol

2. *Find the following limits:*

(a) $\lim_{x \to 1} \frac{\ln x}{x-1}$

Sol

(b) $\lim_{x \to 0} \frac{1-e^{2x}}{x}$

Sol

(c) $\lim_{x \to 0} \frac{e^{ax}-e^{bx}}{x}$

Sol

(d) $\lim_{x \to 0} \frac{e^{-x}-e^{2x}}{x}$

Sol

(e) $\lim_{x \to 0} \frac{e^{-x^2}-1}{x}$

Sol